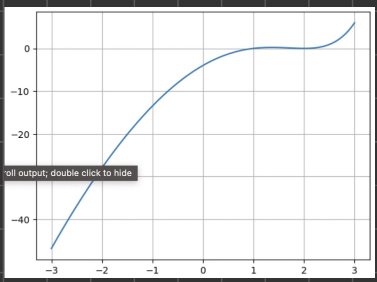
Prob 1 – Bisection Method

$$f(x) = (1+x)\,2^{x+2} + (2^x-2)x^2 + 8(x-1) \quad , \quad x \in [-3,3]$$

a)



b) $a = -3 \;,\; b = 3 \;,\; c = \frac{a+b}{2} = \frac{-3+3}{2} = 0$

$f(0) = (1+0)\,2^{0+2} + (\cancel{2^x-2})0^2 + 8(0-1) = -4$

$f(-3) = -46{,}9$

$f(a)f(c) > 0 \Rightarrow a = c = 0 \;,\; b = b = 3 \;,\; c = \frac{b+c}{2} = \frac{3+0}{2} = \frac{3}{2}$

$f(^3/_2) = 0{,}207$

$$\underbrace{f(a)}_{f(0)=-4} f(c) < 0 \Rightarrow a = a = 0 ,\; b = c = \frac{3}{2} \;,\; c = \frac{a+c}{2} = \frac{3}{4}$$

$f(^3/_4) = -0{,}497$

$f(a)f(c) > 0 \Rightarrow a = c = \frac{3}{4} \;,\; b = b = \frac{3}{2} \;,\; c = \frac{b+c}{2} = \frac{\frac{3}{4}+\frac{3}{2}}{2} = \frac{9}{8}$

$\varepsilon < 0{,}001$?

$f(^9/_8) = 0{,}139$

$f(a)f(c) < 0 \Rightarrow a = a = \frac{3}{4} ,\; b = c = \frac{9}{8} \;,\; c = \frac{\frac{3}{4}+\frac{9}{8}}{2} = \frac{15}{16}$

$f(^{16}/_{15}) = -0{,}0957$

$f(a)f(c) < 0 \Rightarrow a = a = \frac{3}{4} ,\; b = c = \frac{15}{16} \;,\; c = \frac{a+c}{2} = \frac{27}{32}$

$f(^{27}/_{32}) = -0{,}275$

$|e_n| \leq 2^{-(n+1)}(b_0 - a_0)$

$\frac{10^{-3}}{6} \leq 2^{-(n+1)} \qquad -(n+1) = \frac{\log(10^{-3}/6)}{\log(2)} = -12{,}55 \Rightarrow n = 12{,}55 - 1 = 11{,}55 \approx 12$

$6 \cdot 10^3 \geq 2^{n+1} \Rightarrow n \leq \frac{\log(6000)}{\log(2)} - 1$

$|e_n| \leq \frac{(b_0-a_0)/2}{2^n}$

$10^{-3} \leq \frac{3}{2^n} \Rightarrow 10^3 \geq \frac{2^n}{3} \Rightarrow n = \frac{\log(3000)}{\log(2)} = 11{,}5 \Rightarrow n < 12$